

Static vs Contextual Embeddings for Lexical Ambiguity

A Word-in-Context Evaluation

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Introduction

“She sat on the **bank** of the river.” Here, *bank* means the land beside the water.

“She deposited money at the **bank**.” Here, *bank* means a financial institution.

The spelling is identical, but the meaning changes because of the surrounding words. This is **lexical ambiguity**.

We wanted to understand whether a computer model represents the same word differently when its context changes, and whether that difference helps it decide if the meaning is the same or different.

Inspiration

Our experiment follows the **Word-in-Context (WiC)** task introduced by Pilehvar and Camacho-Collados (2019). WiC avoids asking for a dictionary definition. Instead it asks **does the same word have the same meaning in two sentences?**

Camacho-Collados and Pilehvar (2018) also describe a problem with traditional word representations. One stored vector can mix several meanings together. That motivated our comparison between a fixed representation and representations that use context.

Background

A **word embedding** is a list of numbers used to represent a word computationally. We compare two pretrained approaches called **GloVe** and **BERT**.

GloVe learns one stored vector for each word from patterns of words occurring together in large text collections. The vector for *bank* therefore stays the same in a river sentence and a money sentence. This is a **static embedding**.

BERT reads the surrounding sentence before creating the target word representation. Its representation of *bank* can therefore change between those two sentences. This is a **contextual representation**.

The two meanings of *bank* above are **homonyms** because they are unrelated. Related meanings are called **polysemy**. For example, *paper* can mean writing material or a newspaper. The names are a mouthful, but the distinction is simple. It separates unrelated meanings from related ones.

WiC does not ask the model to name a sense. It only asks whether the target word has the **same meaning** or a **different meaning** in the two sentences.

Hypothesis

We expected more sentence context to make changes in meaning easier to recognise.

With GloVe target only, *bank* receives the same vector in both sentences, so the method cannot see the river versus money difference. The validation set has **638 pairs** with 319 same meaning and 319 different meaning. A method that always chooses one answer would still get half correct, so **50% is our simple baseline**.

We expected nearby words such as *river* or *money* to help GloVe somewhat. We expected BERT to do better because it uses the full sentence when representing the target word.

Methodology

The dataset

We used the **SuperGLUE Word-in-Context (WiC)** dataset for every comparison. Each example contains **two sentences with the same marked word** and a label telling us whether that word has the same meaning in both.

A river side *bank* paired with a financial *bank* is labelled **different meaning**. Two financial uses of *bank* would be labelled **same meaning**.

WiC provides **5,428 labelled training pairs** and **638 labelled validation pairs**. We used the training data to choose the settings for each method, then kept those choices fixed and measured final performance on validation.

The release also contains **1,400 test pairs**, but their correct labels are not public. We therefore cannot calculate a local test score and report validation results instead.

The comparison

We used **two pretrained model families, GloVe and BERT, in three experimental conditions**. The dataset stayed identical in all three.

1. GloVe target word only. We use the stored GloVe 6B 300 dimensional vector of the marked word. The same word receives the same vector in both sentences, giving us a context free baseline.

2. GloVe with nearby words. We take up to **two words before and two words after the target**, leave out the target itself, and average those GloVe vectors.

“The **muddy river bank flooded after** rain.”
For *bank*, the context words are **muddy, river, flooded, after**.

3. BERT using the full sentence. We use frozen google-bert/bert-base-uncased. BERT base has 12 processing layers. Our selected configuration combines the target representation from the **last four layers**. We do not fine tune BERT on WiC.

From vectors to an answer

For each sentence pair, a method gives us two target representations. We compare them with **cosine similarity**. A higher score means the representations are more alike. A lower score means they are more different.

Using training data, we select a cutoff for each condition. Scores above the cutoff predict **same meaning**. Scores below it predict **different meaning**. Our main score is **accuracy**, the percentage of the 638 validation pairs answered correctly.

Results

All three conditions were evaluated on the same **638 validation pairs**.

GloVe target only answered **319 pairs correctly (50.0%)**. Adding nearby GloVe words increased this to **354 pairs (55.5%)**. BERT performed best with **428 correct pairs (67.1%)**.

Validation accuracy

GloVe target only		50.0%	(319/638)
GloVe nearby words		55.5%	(354/638)
BERT in full sentence		67.1%	(428/638)

+11.6 percentage points

Compared with nearby word GloVe, BERT answered **74 additional sentence pairs correctly**.

We repeated that comparison on **1,000 resampled versions** of the validation set. The estimated 95% interval for BERT’s advantage was **6.6 to 16.9 percentage points**, so the advantage remained positive throughout this interval.

Looking pair by pair, both stronger methods were correct on **250 pairs**. BERT alone was correct on **178**, nearby word GloVe alone on **104**, and both were wrong on **106**. BERT was stronger on this benchmark, but neither method solved every example.

Conclusion

Our results support the idea that **context helps a model recognise a word’s meaning**. The fixed GloVe vector could not distinguish the two uses of a word. Nearby words helped a little, and BERT’s full sentence representation did better.

BERT still missed **210 of 638 pairs**. We found a clear improvement, not a complete solution. Building and checking this comparison ourselves made the difference between static and contextual embeddings much easier to see.